

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

3/13

Q.1 Write short answer of the following questions: (5x5=30)

- I. Explain with example the following terms in MATLAB
 - a. polyval()
 - b. roots()
 - c. factor()
- II. Write syntax with example for the followings in MATLAB:
 - a. Extrapolation of data
 - b. Curve fitting
- III. Write a MATLAB program with function to calculate $f(x)$, given by:

$$f(x) = 28x^3 - 15x^2 + 7x - 3$$

Calculate, print and plot values for x and $f(x)$. Also find minimum and maximum of $f(x)$ values for fifteen different values of x .

- IV. Write MATLAB to plot 'x' against $f(x)$ values using subplot such that: $x = [0 \ 4\pi]$, $y(x) = n \cdot \sin(x)$ and $n=1,2,3,4$
- V. Write MATLAB program to determine corresponding force values for the work and distance values as give in the table. Also calculate total force and average work values.

w (work)	20	27	47.5	62
d(distance)	12	27	41	56

VI. Two vectors are given:

$$u = 5i + 9j - 2k \text{ and } v = 4i - 6j + 12k$$

Use MATLAB to calculate the dot product $u \cdot v$ of the vectors in two ways:

- a) Define u as row vector and v as a column vector, and then use matrix multiplication.
- b) Use the dot function.

Solve the following.

Q.2. (a) Write a MATLAB program to plot x vs y such that:

$$\psi_n(x) = A \frac{\pi x x}{L}$$

$$\text{Where } A = \sqrt{\frac{2}{L}}, L = 4, n = [1:4]$$

(6 Marks)

(b) Write MATLAB program to evaluate the $\int_0^1 \left(\frac{\sqrt{x}}{6}\right) dx$ by Trapezoidal Rule. (n=6) (4 Marks)

Q.3. Write MATLAB program to describe motion of a charged particles in magnetic field. Also draw estimate output graphs with proper curve labels, x & y labels and title. (10 Marks)

Q.4. Create matrix D in MATLAB such that

$$D = \begin{bmatrix} 10 & 9 & 8 & 7 & 6 \\ 12 & 14 & 16 & 18 & 20 \\ 1 & 4 & 7 & 10 & 13 \\ 25 & 8 & 11 & 2 & 1 \\ 33 & 14 & 29 & 6 & 12 \end{bmatrix}$$

- (i) Write v1 equals 4th column of D.
- (ii) Find location of value 18 in v1, (iii) create a linear vector k containing all elements of D (iv) create v2 equals 3rd row of D. (v) determine $u = v1 \cdot v2$ (vi) plot 4th column against 1st row of D (vii) select and sort in ascending order the 4th and 5th rows of D. (viii) select a 3x3 middle matrix as an out of D (10

Computational-Physics (II)

ADP Spring 2023

Q#01 write short answer of following questions:

~~a- Polyval()~~
a- Polyval():-

⇒ Description:- returns the value of polynomial at point 'x'.

⇒ Syntax:-

⇒ Example:- % let's say, you have a polynomial $3x^2 + 2x + 1$ & evaluate this polynomial at $x=2$.

>> p=[3 2 1];

>> x=2;

>> y=polyval(p,x)

>> disp('evaluate polynomial');

>> disp('y');

b- roots():-

⇒ Description:- roots(c) computes the roots of polynomial whose co-efficients are the elements of the vector c.

⇒ Syntax:-

roots(c).

⇒ Example:-

b- fact

⇒ Dis

facto

retu

exp

⇒ sy

⇒ Ex

a-

% polynomial is $2x^3 - 6x^2 + 2x - 1 = 0$ and find its roots:-

```
>> p = [2 -6 2 1];  
>> r = roots(p);  
>> disp('roots of polynomial:');  
>> disp(r);
```

b- factor():-

⇒ Description:- $F = \text{factor}(x)$, returns all irreducible factors of x in vector F . If x is an integer, factor returns the prime factorization of x . If x is a symbolic expression, factor returns subexpressions that are factors of x .

⇒ syntax:-

$F = \text{factor}(x)$

⇒ Example:-

% find the prime factor of number 60.

```
>> n = 60;  
>> f = factor(n);  
>> disp('Factor of n:');  
>> disp(f);
```

(II) b

a- Extrapolation of data:-

⇒ Descriptions:-

Extrapolation is the process of estimating unknown values by extending a known set of data.

here, x & y are known data points

x_i = points at which to extrapolate.

method:- linear, nearest, spline, pchip.

'extrap': - specifies that extrapolation should be used for points outside range of x .

⇒ Syntax:-

$y_i = \text{interp1}(x, y, x_i, \text{method}, 'extrap')$

⇒ Example:- 1. Take x & y from user and extrapolate the value at $x=6$.

% known data process:-

>> $x = [1 \ 2 \ 3 \ 4 \ 5];$

>> $y = [2 \ 4 \ 6 \ 8 \ 10];$

>> $x_i = 6;$

>> $y_i = \text{interp1}(x, y, x_i, 'linear', 'extrap');$

>> $\text{disp}(y_i);$

b- Curve fitting:-

⇒ Description:- Curve fitting is the process of constructing a curve - or mathematical function that has the best fit to a series of data points.

⇒ Syntax:- `fitresult = fit(x, y, fitType)`

x: The independent variable data (vector).

y: The dependent variable data (vector)

fitType: A string specifying type of model to fit (e.g., 'poly1' for a linear fit, 'poly2' for a quadratic fit).

⇒ Example:- Taking x & y from user that appears to have a linear relationship, and fit a straight line to it.

% example data

>> `x = [1 2 3 4 5];`

>> `y = [2.2 4.1 5.9 8.2 10.1];`

>> `fitresult = fit(x, y, 'poly1');`

>> `disp(fitresult);`

% plot data & the fit

>> `plot(fitresult, x, y);`

>> `xlabel('x');`

>> `ylabel('y');`

>> `title('Linear fit');`

>> `legend('Data', 'Fitted curve');`

~~d.III~~

% Define given data

>> `x = [-10:15:10];`

>> `f(x) = 28*x.^3 - 15*x.^2 + 7*x - 3;`

>> y(x) = polyval(f(x), x);

% plot data

>> plot(y(x));

>> xlabel('values of x');

>> ylabel('value of polynomial');

>> title('plot of y(x)');

>> grid on;

% minimum of f(x) for x:

>> [min-y, min-index] = min(y(x));

>> disp('min of y(x):');

>> disp(min-y);

>> disp(min-index);

% maximum of f(x) for x:

>> [max-y, max-index] = max(y(x));

>> disp('max of y(x):');

>> disp(max-y);

>> disp(max-index);

end

~~dV~~

% define given data

>> W = [20, 27, 47.5, 62];

>> d = [12, 27, 41, 56];

% Find force

>> F = W./d;

% Find Total-Force

>> Total-Force = sum(F);

% Find avg-work

>> avg-work = mean(W);

% display results

>> disp('Force:');

>> disp(F);

>> disp('Sum of Force:');

>> disp('Total-Force');

>> disp('average of work?');
>> disp(avg-work);

~~-(v1)~~

% define given data

>> u = [5, 9, -2];
>> v = [4; -6; 12];

% Dot product by matrix multiplication

>> Dot-product = u.v;
>> disp('Dot product of u & v');
>> disp(Dot-product);

% Dot product by dot function

>> Dot-p = dot(u, v);
>> disp('Dot product by dot function');
>> disp(Dot-p);

Solve the following

Q#2:- ~~(a)~~

% define given data

```
>> y = (A * n * pi * x) / L;  
>> x = linspace(-10, 10, 4);  
>> L = 4;  
>> A = sqrt(2/L);
```

% plot data

```
>> plot(x, y, '---o');  
>> xlabel('x-axis');  
>> ylabel('y-axis');  
>> title('plot x vs y');  
>> grid on;  
end
```

~~(b)~~

% Trapezoidal method

```

>>  $y = \sqrt{x}/6$ ;
>> a = 1;
>> b = 6;
>> Trapezoidal-y = Trapz(y, x, a, b);
>> disp('Trapezoidal wrt x');
>> disp(Trapezoidal-y);
end

```

Q#3:- % given data.

```

>> q = 1; >> m = 1; >> h = 0.5; >> tmax = 100;
>> x = 100; >> y = 20; >> V0 = 5; B = 0.5;
>> w = (q * B) / m;
>> ind = 1;
>> t = 1;
while (t <= tmax)
>> ta(ind) = t;
>> xa(ind) = x;
>> ya(ind) = y;
>> Vx = V0 * sin(w * t);
>> Vy = V0 * cos(w * t);
>> ax = w * Vy;
>> ay = -w * Vx;
>> Vx = Vx + ax * h; >> Vy = Vy + ay * h;
>> x = x + Vx * h; >> y = y + Vy * h;
>> t = t + h; >> ind = ind + 1;
end

```


% display result

```
>> disp('time x y');  
>> tb(:, 1) = ta';  
>> tb(:, 2) = xa';  
>> tb(:, 3) = ya';  
>> disp(tb);
```

% plot & label ta, xa

```
>> plot(ta, xa, '--o');  
>> hold on;  
>> xlabel('time');  
>> ylabel('position-x');  
>> title('plot of ta vs xa');  
>> grid on;
```

% plot & label ta, ya.

```
>> plot ( ta, ya, '-o' );  
>> x label ('time');  
>> y label ('position-y');  
>> title ('plot of ta vs ya');  
>> grid on;  
>> legend ('Time , position-x, position-y');  
end
```

Q#4:-

% create matrix D as given

```
>> D = [ 10, 9, 8, 7, 6; 12, 14, 16, 18, 20; 1, 4, 7, 10,  
        13; 25, 8, 11, 2, 1; 33, 14, 29, 6, 12];
```

(a) % create V1 equal to 4th column of D

```
>> V1 = D(:, 4);
```

(b) % location of value 18 in v1

```
>> index = find (V1 == 18);
```


(c) % create a linear vector k containing all elements of D

>> k = D(:);

(d) % create v2 equal to 3rd row of D

>> v2 = D(3,:);

(e) % Determine u

>> u = dot(v1, v2);

(f) % plot 4th column against 1st row of D.

>> plot(D(:,4), D(1,:));

>> xlabel('4th column of D');

>> ylabel('1st row of D');

>> title('plot of 4th column against 1st row');

>> grid on;

(g) % select & sort in ascending order

>> sorted-rows = sort(D([4,5],:), 2);

(h) % 3 x 3 middle matrix

>> middle-matrix = D(2:4, 2:4);

~~(IV)~~

% define given data

```
>> x = [0 : pi/10 : 4*pi]
>> for n = 1:4;
>> y = sin(n*x);
>> subplot (1,4,n);
>> plot ( x,y);
>> xlabel('x');
>> ylabel('y');
>> title('plot x vs y');
end
```